

# **Lecture Notes**

## **EC226 Term 1 2025/26**

### **Errors in Variables**

#### **Readings:**

- Stock and Watson (2003): Chapter 9.2
- Dougherty (2016): Chapter 8
- Wooldridge (2013): Chapter 9

## Contents

|          |                                            |          |
|----------|--------------------------------------------|----------|
| <b>1</b> | <b>Measurement error in <math>x</math></b> | <b>3</b> |
| <b>2</b> | <b>Measurement error in <math>y</math></b> | <b>4</b> |

# 1 Measurement error in $x$

Consider the case

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i \quad (1)$$

where the error term  $\varepsilon_i$  satisfies the CLRM assumptions, such that  $\varepsilon_i | x \sim N(0, \sigma^2)$ . However, we do NOT observe  $x$  directly, but observe  $x^*$ , which is measured with error, that is:

$$x_i^* = x_i + \eta_i$$

where  $x_i$  is the actual true value, and  $x_i^*$  is the recorded value and  $\varepsilon_i \sim N(0, \sigma_\eta^2)$ .

Substituting out  $x_i$  in equation (1) using the recorded value  $x_i^*$  we obtain:

$$y_i = \beta_0 + \beta_1 x_i^* + u_i + \beta_1 (x_i - x_i^*) = \beta_0 + \beta_1 x_i^* + \underbrace{\varepsilon_i - \beta_1 \eta_i}_{v_i} \quad (2)$$

OLS estimation of equation (2) yields the equation for the slope coefficient as:

$$b_1 = \frac{\sum_i (x_i^* - \bar{x}^*)(y_i - \bar{y})}{\sum_i (x_i^* - \bar{x}^*)^2} \quad (3)$$

substituting out  $(y_i - \bar{y})$  in equation (3) using equation (2) we have:

$$\begin{aligned} E(b_1) &= \beta_1 + \frac{\sum_i (x_i^* - \bar{x}^*) v_i}{\sum_i (x_i^* - \bar{x}^*)^2} \\ &= \beta_1 + \frac{\text{cov}(x_i^*, v_i)}{\text{var}(x_i^*)} \\ &= \beta_1 + \text{cov}(x_i^*, v_i) / \text{var}(x_i^*) \\ &= \beta_1 + \text{cov}(x_i^*, \varepsilon_i - \beta_1 \eta_i) / \text{var}(x_i^*) \\ &= \beta_1 + \text{cov}(x_i^*, \varepsilon_i) / \text{var}(x_i^*) - \beta_1 \text{cov}(x_i^*, \eta_i) / \text{var}(x_i^*) \\ &= \beta_1 - \beta_1 \sigma_\eta^2 / \text{var}(x_i^*) \\ &= \beta_1 \left( 1 - \frac{\sigma_\eta^2}{\sigma_x^2 + \sigma_\eta^2} \right) \\ &= \beta_1 \left( \frac{\sigma_x^2}{\sigma_x^2 + \sigma_\eta^2} \right) \end{aligned} \quad (4)$$

This hold as  $\text{cov}(x_i^*, \text{varepsilon}_{\varepsilon_i}) = 0$ ,  $\text{cov}(x_i^*, \eta_i) = \text{cov}(x_i + \eta_i, \eta_i) = \underbrace{\text{cov}(x_i, \eta_i)}_0 + \text{cov}(\eta_i, \eta_i) = \sigma_\eta^2$ ,

and  $\text{var}(x^*) = \sigma_x^2 + \sigma_\eta^2$ . Therefore as  $0 < \left( \frac{\sigma_x^2}{\sigma_x^2 + \sigma_\eta^2} \right) < 1$  we know  $E(b_1) < \beta_1$ , i.e. the OLS estimator is biased towards zero (irrespective of the sign of  $\beta_1$ ).

## 2 Measurement error in $y$

Consider instead the case when

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i \quad (5)$$

where the error term  $\varepsilon_i$  satisfies the CLRM assumptions, but  $y_i^* = y_i + \eta_i$ , where  $y_i$  is the actual value,  $y_i^*$  is the recorded value and  $\eta_i \sim N(0, \sigma_\eta^2)$ . Now, on substituting out  $y_i$  using the recorded value  $y_i^*$  in equation (5) we obtain

$$y_i^* - \eta_i = \beta_0 + \beta_1 x_i + \varepsilon_i \Rightarrow y_i^* = \beta_0 + \beta_1 x_i + \underbrace{\varepsilon_i + \eta_i}_{v_i} \quad (6)$$

as  $v_i$  has  $E(v_i) = 0$ ,  $var(v_i) = \sigma^2 + \sigma_\eta^2$  and  $cov(v_i, v_j) = 0$ ,  $i \neq j$ , then OLS will yield unbiased and consistent estimates.

## References

Dougherty, C. (2016). *Introduction to Econometrics*. OUP Catalogue. Oxford University Press.

Stock, J. and Watson, M. W. (2003). *Introduction to Econometrics*. Prentice Hall, New York.

Wooldridge, J. M. (2013). *Introduction to Econometrics*. Cengage.